

# **MINI PROJECT PHASE 2 REPORT**

Name

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Project Title

**Sales Transaction Analysis and Business Insights**

Power BI Visualization and Interactive Dashboard

Tools Used: Microsoft Power BI, Power Query and DAX

Dashboard Development and Visualization

## 1. Introduction

Phase 2 focused on converting the prepared sales transaction data into an interactive Power BI dashboard. In Phase 1, the data was cleaned and organized into fact and dimension tables. In this phase, I used that prepared structure to create visualizations that make sales patterns easier to understand. The work included creating measures, designing dashboard pages, adding filters and navigation, and making the report easier to explore.

## 2. Power BI Report Setup

The cleaned tables were loaded into Power BI and checked before visualization. The existing star-schema structure was retained so the relationships between the fact table and dimension tables could be used directly for analysis.

| Table             | Purpose                                           |
|-------------------|---------------------------------------------------|
| Fact_Transactions | Transaction-level sales, quantity, dates and keys |
| Dim_Customer      | Customer information and customer_key             |
| Dim_Product       | Product categories and product_key                |
| Dim_Country       | Country information and country_key               |
| Dim_Payment       | Payment methods and payment_key                   |
| Dim_Date          | Date attributes for time-based analysis           |

## 3. DAX Measures

I created explicit DAX measures for the main business metrics used throughout the dashboard. Using measures keeps the calculations consistent and allows them to respond automatically to slicers and filters.

| Measure             | Purpose                                 |
|---------------------|-----------------------------------------|
| Total Sales         | Displays the overall sales amount       |
| Total Transactions  | Displays the number of transactions     |
| Total Quantity      | Displays the total quantity sold        |
| Average Order Value | Shows the average value per transaction |
| Average Unit Price  | Shows the average price per unit        |

## 4. Visualization Approach

I selected visuals based on what each part of the data needed to communicate. KPI cards were used for quick summaries, charts for comparisons and trends, and slicers for interactive filtering.

| Visual             | Purpose                                        |
|--------------------|------------------------------------------------|
| KPI Cards          | Quick view of the main business measures       |
| Line Chart         | Show changes in sales over time                |
| Bar / Column Chart | Compare categories, countries and other groups |
| Donut / Pie Chart  | Show payment-method distribution               |
| Slicers            | Let users filter the report interactively      |

## 5. Dashboard Pages

The report was divided into focused pages so each section has a clear purpose. I kept the same theme and layout style across the pages so the dashboard feels like one complete report.

### 5.1 Home Page

The Home page was created as the starting point of the report. It contains the project title and navigation to the main dashboard sections. I used a modern purple-themed design with a dark background and glass-style elements to give the report a more polished appearance.

### 5.2 Sales Overview

The Sales Overview page provides the main summary of the dataset. KPI cards show Total Sales, Total Transactions, Total Quantity, Average Order Value and Average Unit Price. Supporting visuals show sales trends, product categories, countries and payment methods.

### 5.3 Product & Country Analysis

This page compares sales across product categories and countries. It helps identify stronger categories and understand how sales are distributed geographically.

### 5.4 Time & Trend Analysis

This page uses the Dim\_Date table to visualize sales across months, quarters and years. It provides a clearer view of how sales change over time.

### 5.5 Customer & Payment Analysis

This page focuses on customer activity and payment preferences, providing a more detailed view of who is involved in the transactions and how purchases are paid for.

## 6. Slicers and Interactivity

Slicers were added so users can interact with the dashboard instead of viewing only fixed charts. Selecting a value updates the related visuals and makes it possible to explore different parts of the dataset.

| Feature            | Use                                                |
|--------------------|----------------------------------------------------|
| Year               | Filter analysis by year                            |
| Country            | Focus on a selected country                        |
| Payment Method     | Filter and compare payment methods                 |
| Product Category   | Focus on selected categories                       |
| Visual Interaction | Use selections in charts to filter related visuals |

## 7. Page Navigation

A page navigator was added to make movement between the main report pages simple. The Home page acts as the entry point, while the analytical pages contain the detailed visualizations. Navigation was tested after setup to make sure the intended pages open correctly.

## 8. Drill-Through Visualization

Drill-through was added to provide detail without overcrowding the main dashboard. A user can select a relevant customer or product and open a dedicated detail page while keeping the selected context.

| Drill-through Page             | Purpose                         | Field            |
|--------------------------------|---------------------------------|------------------|
| Customer Details_Drill Through | Focused customer-level analysis | customer_name    |
| Product Details_Drill Through  | Focused product-level analysis  | product_category |

### 8.1 Back Navigation

A back button was added to the drill-through pages so the user can return to the page from which the detailed view was opened.

## 9. Dashboard Formatting and Theme

The dashboard uses a consistent purple-based theme, aligned visual containers and clear headings. The pages were designed in a 16:9 layout for a standard widescreen dashboard. Formatting was applied consistently to KPI cards, charts, slicers, titles and navigation elements.

## 10. Testing and Validation

After completing the report, I tested the navigation, slicers, visual interactions and drill-through functionality. I also checked that filters affected the expected visuals and that the overall formatting remained consistent and readable.

## 11. Conclusion

In Phase 2, I transformed the prepared sales data into an interactive Power BI visualization report. The final dashboard combines DAX measures, charts, slicers, navigation and drill-through features in a consistent design, making the transaction data easier to explore and understand.